

MATLAB CODE FOR APC-AXIN- β -CATENIN MODEL

For your first task, I want you to create a well written Matlab file that outputs a trajectory of the APC-Axin- β -catenin model from the 2017 paper titled "A kinetic model to study the regulation of β -catenin, APC, and Axin in the human colonic crypt." This paper is in the GitHub folder. We will use the code that you make later in the PDE model. Here are some notes:

- 1.] I do not want you to recreate the entire model using all 17 equations, I only want you to model the *long-term dynamics* of the system, which are given in Equations 102, 103, and 104. This will be a three-dimensional, very nonlinear, system of equations.
- 2.] In the paper, X_7 is APC; X_{11} is β -catenin; and X_{12} is Axin. Change the notation so that APC is $A(t)$; β -catenin is $B(t)$; and Axin is $X(t)$.
- 3.] Your code will likely start be defining all the parameters, which are given in Table 2.
- 4.] The Wnt signal, given by W , is an input scalar (essentially a parameter) into the system.
- 5.] Notice that you'll have to define the two functions f and g in order to define the three functions F , G , and H as function handles, which then will allow you to define the right-hand side of each differential equation. Below, I show you the dependence of each function:

$$\begin{aligned}
 f &:= f(W) \\
 g &:= g(A, B, X, W) \\
 F &:= F(A, B) \\
 G &:= G(A, B, X, W) \\
 H &:= H(A, B, X)
 \end{aligned}$$

- 6.] Once you have all of those functions defined, you can then define $\mathbf{dUdt} = @(\mathbf{t}, \mathbf{U}) \dots$ as the vector of right-hand side functions using A , B , X as $U(1)$, $U(2)$, and $U(3)$, respectively.
- 7.] The goal is to output the same phase portrait plot as in Figure 5 from the paper. The main input is $W = 1$ with the output being that figure. You do not have to plot the *null surfaces*.